RIPHAH
INTERNATIONAL UNIVERSITY

Name : Muhammad Ammanullah

Sap I`d : 55633

Section : BS SE (5 – 2)

Course : Artificial Intelligence

Project

Machine Learning Project Report

Comparative Analysis of Machine Learning Models for Distance Prediction

Project Overview

This project involves the application of machine learning algorithms to analyze and compare the performance of various technologies. The goal is to identify and optimize a high-performing algorithm for the dataset under study, ultimately presenting a comprehensive analysis of the results.

1. Introduction

This report presents a comprehensive evaluation of multiple machine learning models using dataset features to predict distances based on RSSI values. The models evaluated include Logistic Regression, Decision Tree, Random Forest, SVM, KNN, Neural Network, and Naive Bayes. Metrics such as Accuracy, Precision, Recall, F1 Score, and Cross-Validation scores were computed for a comparative analysis.

2. Dataset Description

The dataset consists of signal strength readings (RSSI) and corresponding distance labels. Features and target variables used in the analysis are as follows:
- Features: RSSI, Node
- Target: Distance

3. Preprocessing Steps

i. Data Splitting

The dataset was split into:

- **Training Set**: 80%
- **Test Set**: 20%

ii. Feature Scaling

- **Scaler**: Standard Scaler was applied to ensure uniform scaling for algorithms sensitive to feature magnitudes.

iii. Stratified Cross-Validation

- A 5-fold Stratified K-Fold Cross-Validation was used to ensure balanced class distributions across folds.

3. Methodology

The following methodology was used:-

1. **Data Preprocessing:** Standardized the features for models requiring scaled inputs.
2. **Train-Test Split:** Dataset split into 80% training and 20% testing using stratified sampling.
3. **Model Evaluation:** Trained models and computed metrics including accuracy, precision, recall, F1 score, and cross-validation scores.
4. **Hyperparameter Tuning:** Performed Grid Search to optimize hyperparameters for selected models.

Machine Learning Models

The following models were evaluated:

1. Logistic Regression
2. Decision Tree Classifier
3. Random Forest Classifier
4. Support Vector Machine (SVM)
5. K-Nearest Neighbors (KNN)
6. Neural Network
7. Naive Bayes

4. Results

Below are the evaluation results of the machine learning models. Metrics were calculated based on predictions on the test set.

BLE Technology Performance

Results Table:

Model	Accuracy	Precision	Recall	F1 Score	Cross-Validation
Logistic Regression	0.464945	0.454406	0.464945	0.458384	0.447538
Decision Tree	0.645756	0.651114	0.645756	0.647024	0.615569
Random Forest	0.505535	0.451296	0.505535	0.415622	0.49586
SVM	0.566421	0.657521	0.566421	0.567692	0.544349
KNN	0.607011	0.609211	0.607011	0.600548	0.594743
Neural Network	0.54428	0.587351	0.54428	0.551627	0.568337
Naive Bayes	0.448339	0.42117	0.448339	0.428867	0.452029

WIFI Technology Performance

Results Table:

Model	Accuracy	Precision	Recall	F1 Score	Cross-Validation
Logistic Regression	0.610727	0.605755	0.610727	0.60684	0.612828
Decision Tree	0.543253	0.42776	0.543253	0.451276	0.472531
Random Forest	0.543253	0.42776	0.543253	0.451276	0.535906
SVM	0.66436	0.677951	0.66436	0.659054	0.67071
KNN	0.653979	0.653997	0.653979	0.653137	0.67409
Neural Network	0.653979	0.652253	0.653979	0.649561	0.676349
Naive Bayes	0.591696	0.589823	0.591696	0.587324	0.585502

ZIGBEE Technology Performance

Results Table:

Model	Accuracy	Precision	Recall	F1 Score	Cross-Validation
Logistic Regression	0.494755	0.471183	0.494755	0.478937	0.50399
Decision Tree	0.875874	0.889478	0.875874	0.875945	0.831657
Random Forest	0.912587	0.916328	0.912587	0.912147	0.901957
SVM	0.743007	0.752416	0.743007	0.741583	0.740449
KNN	0.882867	0.887328	0.882867	0.882628	0.88245
Neural Network	0.840909	0.843561	0.840909	0.838048	0.843482
Naive Bayes	0.494755	0.471183	0.494755	0.478937	0.509257

LORAWAN Technology Performance

Results Table:

Model	Accuracy	Precision	Recall	F1 Score	Cross-Validation
Logistic Regression	0.737847	0.741371	0.737847	0.739422	0.772653
Decision Tree	0.753472	0.771944	0.753472	0.749802	0.783137
Random Forest	0.829861	0.862455	0.829861	0.83242	0.862952
SVM	0.868056	0.868706	0.868056	0.867101	0.870896
KNN	0.904514	0.909224	0.904514	0.905038	0.908964
Neural Network	0.914931	0.91651	0.914931	0.915338	0.914964
Naive Bayes	0.654514	0.630387	0.654514	0.635944	0.666813

Model Performance Before and After Hyperparameter Tuning

Results Table:

Technology	Model	Test Accuracy (Before Tuning)	Test Accuracy (After Tuning)
Zigbee	Decision Tree	0.875874	0.909091
	KNN	0.882867	0.902098
	Logistic Regression	0.494755	0.54021
	Naive Bayes	0.494755	0.494755
	Neural Network	0.840909	0.81993
	Random Forest	0.912587	0.912587
	SVM	0.743007	0.816434

Results Table:

Technology	Model	Test Accuracy (Before Tuning)	Test Accuracy (After Tuning)
BLE	Decision Tree	0.645756	0.654982
	KNN	0.607011	0.607011
	Logistic Regression	0.464945	0.446494
	Naive Bayes	0.448339	0.448339
	Neural Network	0.54428	0.54428
	Random Forest	0.505535	0.614391
	SVM	0.566421	0.581181

Results Table:

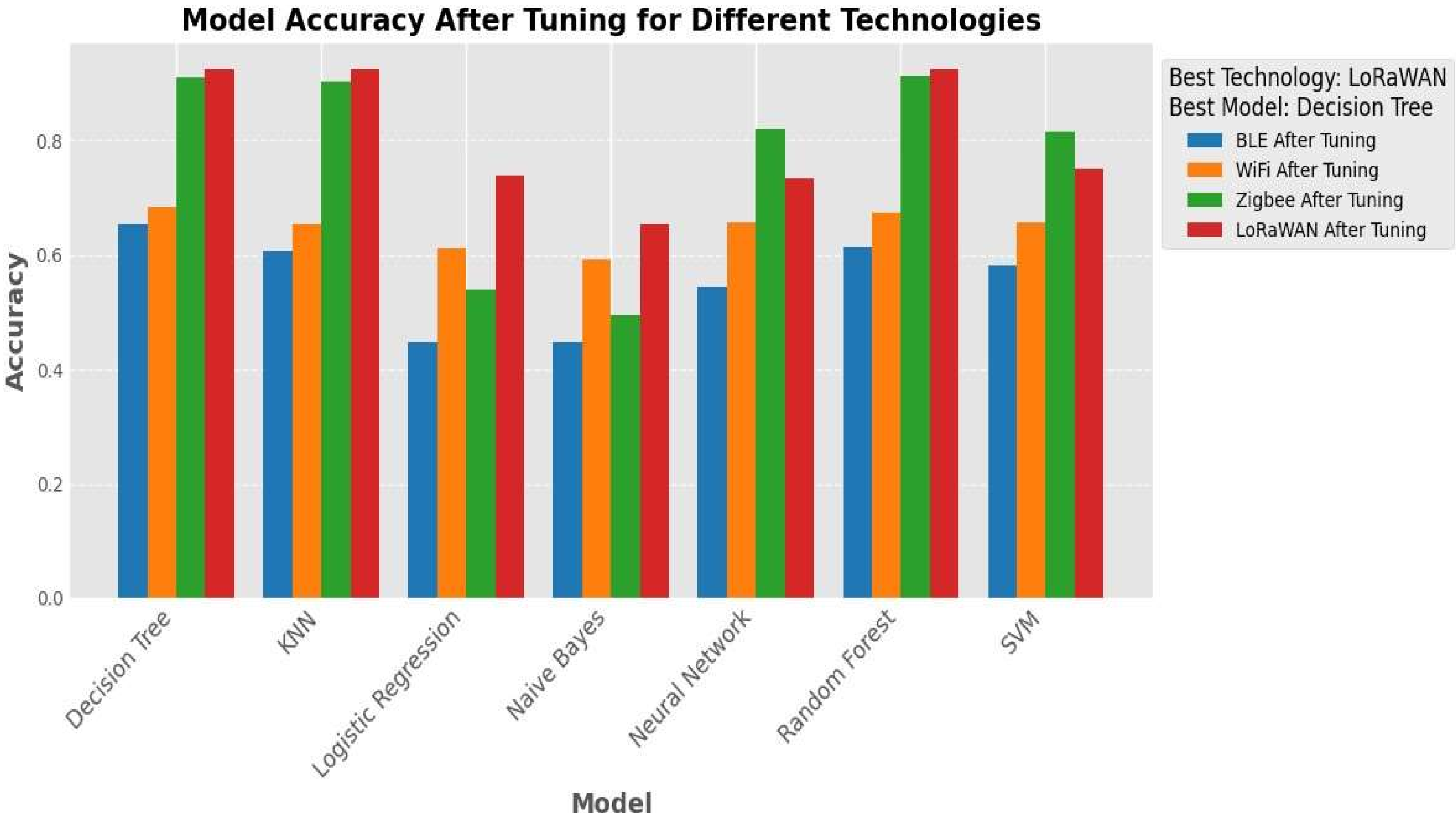
Technology	Model	Test Accuracy (Before Tuning)	Test Accuracy (After Tuning)
WiFi	Decision Tree	0.543253	0.683391
	KNN	0.653979	0.653979
	Logistic Regression	0.610727	0.610727
	Naive Bayes	0.591696	0.591696
	Neural Network	0.653979	0.655709
	Random Forest	0.543253	0.67301
	SVM	0.66436	0.657439

Results Table:

Technology	Model	Test Accuracy (Before Tuning)	Test Accuracy (After Tuning)
LoRaWAN	Decision Tree	0.753472	0.925347
	KNN	0.923611	0.925347
	Logistic Regression	0.737847	0.737847
	Naive Bayes	0.654514	0.654514
	Neural Network	0.744792	0.734375
	Random Forest	0.829861	0.925347
	SVM	0.671875	0.751736

5. Visualization and Final Results

The bar plot offers a visual comparison of the models' performance across different technologies.



6. Conclusion

This analysis demonstrates the varying performance of different machine learning models. **Decision Tree** and **Random Forest** outperformed other models in most metrics, making it a suitable choice for the given dataset and problem. Further tuning and feature engineering could enhance model performance.

Git Link

https://github.com/MuhammadAmmanullah/ARTIFICIAL-Intelligence

Code.Ipynb File Link

https://colab.research.google.com/drive/1KPiViPowzJ5RDf0j74Zp3BBeVpQiE09m?usp=sharing

Dataset Link

https://archive.ics.uci.edu/dataset/586/ble+rssi+dataset+for+indoor+localization